**Q.1 In fuzzy sets, the value of the membership function μ(x) lies in:**

**A. [0, 1]**

B. (−∞, ∞)

C. [−1, 0]

D. {0, 1}

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**Q.2 A fuzzy set can be written as:**

A. A = {μA(x)}

B. A = {x, y | x □ μ(y)}

C. A = {x | x □ U}

**D. A = {(x, μA(x)) | x □ U}**

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**Q.3 In classical (crisp) sets, the membership function can take which values?**

A. Between −1 and 1

B. Between 0 and 10

C. Any real number

**D. Only 0 or 1**

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**Q.4 How many elements are there in the power set of a set containing 3 elements?**

**A. 8**

B. 6

C. 9

D. 3

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**Q.5 Which of the following is the correct definition of a set?**

A. A group of numbers only

B. A group of identical objects only

**C. A collection of well-defined and distinct objects**

D. A list of random numbers

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**Q.6 What does μA(x) = 0 mean?**

**A. x does not belong to set A**

B. None of these

C. x partially belongs to set A

D. x fully belongs to set A

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**Q.7 A membership function represents:**

**A. The degree of belongingness of an element to a set**

B. The power of a fuzzy set

C. The intersection of two sets

D. The total number of elements in a set

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**Q.8 The number of elements in the power set of a set with n elements is given by:**

A. n!

B. n

**C. $2^n$**

D. $n^2$

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**Q.9 Which of the following is not a valid membership value?**

**A. 1.5**

B. 1

C. 0.4

D. 0

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**Q.10 The union of two fuzzy sets A and B is defined as:**

**A. μA□B(x) = max(μA(x), μB(x))**

B. μA□B(x) = μA(x) * μB(x)

C. μA□B(x) = min(μA(x), μB(x))

D. μA□B(x) = μA(x) + μB(x)

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